PAPER_413 – Ug3 CCW/CW Differential Rotation: SCm Planetary Core Disk Penetration Framework

***Session:** 110 (grokshare755feea7.txt analysis)*

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Source: grok*share*755feea7.txt — "Star Magic" Chapter 5 & CelestialBody compute*Ug3 sections* ***Session:*** *110 (grok*share_755feea7.txt analysis) **CP4 Class:** `Ug3CCWCWDifferentialRotationSCmPlanetaryCoreCalculator` (#63)

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1. Overview

PAPER_413 derives the **Ug3 magnetic strings disk** arising from the CCW/CW differential rotation on the solar surface and corona. This differential rotation creates a disk of spinning magnetic strings that penetrates **planetary cores** exclusively through the SCm+UA pathway. Externally, Ug3 remains non-interactive with standard matter. The heliospheric current sheet tilt (0°–30°) governs the disk projection angle.

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2. Differential Rotation Mechanism

2.1 Equatorial CCW vs Coronal CW Rotation

The solar surface exhibits **differential rotation** — the equator rotates faster than higher latitudes, and the **coronal plasma rotates in the opposite sense (CW)** at high latitudes:

Region	Direction	Angular velocity
Equatorial surface	CCW (prograde)	$\omega_{s,\text{eq}} = 2.9 \times 10^{-6}$ rad/s
Polar / Coronal	CW (retrograde)	$\omega_{s,\text{pol}} = 2.1 \times 10^{-6}$ rad/s
Weighted average	Mixed	$\omega_{s,\text{avg}} \approx 2.5 \times 10^{-6}$ rad/s

2.2 Solar Cycle Modulation

Differential rotation is modulated by the **11-year solar cycle**:

$$\omega_s(t) = 2.5 \times 10^{-6} - 0.4 \times 10^{-6} \cdot \sin(\omega_c \cdot t) \quad [\text{rad/s}]$$

where $\omega_c = \frac{2\pi}{3.96 \times 10^8}$ s^{-1} is the solar cycle frequency.

2.3 Heliospheric Current Sheet Tilt

The counter-rotating regions create a **heliospheric current sheet** with:

Minimum tilt: $0°$ (solar minimum)

Maximum tilt: $30°$ (solar maximum)

Average: $\theta_c \approx 15°$

This tilt determines the spatial extent of Ug3's penetration reach in the planetary plane.

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3. Ug3 Equation — Full Derivation

$$Ug_3 = k_3 \cdot \sum_j B_j(r, \theta, t, [\text{SCm}]) \cdot \cos\!\left(\omega_s(t) \cdot t \cdot \pi\right) \cdot P_{\text{core}} \cdot E_{\text{react}}$$

Component breakdown:

Symbol	Meaning	Value
k_3	Ug3 coupling constant	1.8 (calibrated)
B_j	Magnetic field of j-th string	$B_j \approx B_s + B_{\text{SCm}} = 10^{-4} + 10^3$ T
$\cos(\omega_s(t) \cdot t \cdot \pi)$	CCW/CW phase modulation	oscillates ±1
P_{core}	Planetary core exclusivity factor	10^{-3}
E_{react}	Reactor efficiency factor	$\approx 10^{46} \cdot e^{-0.0005t}$

3.1 Numerical Ug3 Value

At $t = 0$ with $B_j = 10^3$ T (SCm dominant), $r = R_s$, and 1 representative string:

$$Ug_3(t=0) \approx 1.8 \cdot 10^3 \cdot 1 \cdot 10^{-3} \cdot 10^{46} \approx 1.8 \times 10^{46}$$

With solar cycle variation:

$$Ug_3(t) \approx \left[10^3 + 0.4 \cdot \sin(\omega_c t)\right] \cdot \cos\!\left(\omega_s(t) \cdot t \cdot \pi\right) \cdot e^{-0.0005t}$$

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4. Planetary Core Exclusivity — P_core

4.1 Physical Basis

When a planet is **donated** SCm by its host star during system formation, a quantity $[\text{SCm}]_{\text{planet}}$ is bound within the planetary core. This creates the only pathway for Ug3 to interact with the planet:

$$P_{\text{core}} = \frac{[\text{SCm}]_{\text{planet}}}{[\text{SCm}]_{\text{star}}} = \frac{10^{12} \text{ kg/m}^3}{10^{15} \text{ kg/m}^3} = 10^{-3}$$

4.2 External Non-Interaction Rule

Outside the planetary core (i.e., in the mantle, surface, or atmosphere), Ug3 has **zero interaction** with standard matter:

$$Ug_3^{\text{external}} = 0 \quad \text{(no SCm available)}$$

The core itself contains the exclusive SCm+UA vector field:

$$\vec{F}_{Ug3,\text{core}} = k_3 \cdot B_j \cdot P_{\text{core}} \cdot \hat{r}_{\text{core}} \cdot E_{\text{react}}$$

4.3 Core SCm Density Estimates by Planet

Planet	Est. $\rho_{\text{SCm,core}}$	P_{core}
Sun (reference)	10^{15} kg/m^3	1
Earth	$\sim 10^{12}$ kg/m^3	10^{-3}
Jupiter	$\sim 5 \times 10^{12}$ kg/m^3	5×10^{-3}
Neptune	$\sim 10^{11}$ kg/m^3	10^{-4}

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5. Ug3 Speed: Faster Than All Planets

The Ug3 magnetic string disk propagates at a velocity governed by ω_s:

$$v_{Ug3} = \omega_s \cdot r_{\text{orbital}} \approx 2.5 \times 10^{-6} \times 1.496 \times 10^{11} \approx 374 \text{ m/s}$$

At $r = 40$ AU (Neptunian orbit):

$$v_{Ug3} \approx 2.5 \times 10^{-6} \times 6.0 \times 10^{12} \approx 15000 \text{ m/s}$$

Earth's orbital velocity: $\sim 29{,}784$ m/s — Ug3 disk is **slower** at 1 AU but serves as a standing-wave coupling rather than a particle velocity. The disk is **always present** across all orbital radii simultaneously.

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6. Code: compute_Ug3

```
// CelestialBody.cpp — Ug3 with CCW/CW differential
double compute_Ug3(const CelestialBody& body, double r, double theta, double t, double
tn,
double k3, double rho_A, double kappa) {
double omega_c = 2.0 * M_PI / 3.96e8; // 11-yr solar cycle
double omega_s = 2.5e-6 - 0.4e-6 * sin(omega_c * t);
double Bs_t = 1e-4 + 0.4e-4 * sin(omega_c * t); // sunspot modulation
double B_SCm = 1e3; // SCm magnetic contribution
double Bj = Bs_t + B_SCm; // total B per string
double cos_mod = cos(omega_s * t * M_PI);
double Pcore = body.Pcore; // = 1e-3 for Earth
double Ereact = compute_Ereact(t, body.SCm_density, 1e8, rho_A, kappa);
return k3 * Bj * cos_mod * Pcore * Ereact;
}
```

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7. Unit Tests

```
import math
def test_omega_s_solar_cycle():
"""Differential rotation bounded within [2.1e-6, 2.9e-6] rad/s"""
omega_c = 2 * math.pi / 3.96e8
```

```
for t_yr in range(0, 12):
t = t_yr * 3.156e7
omega_s = 2.5e-6 - 0.4e-6 * math.sin(omega_c * t)
assert 2.1e-6 <= omega_s <= 2.9e-6
def test_Pcore_exclusivity():
"""Pcore = 1e-3 verifies planetary vs stellar SCm ratio"""
rho_SCm_planet = 1e12; rho_SCm_star = 1e15
Pcore = rho_SCm_planet / rho_SCm_star
assert abs(Pcore - 1e-3) < 1e-20
```
